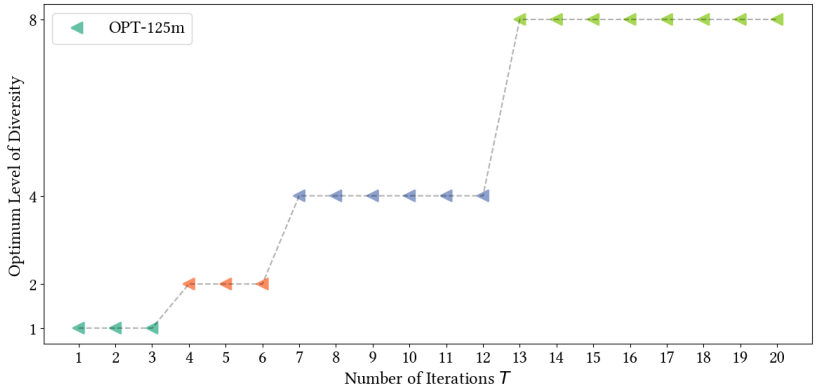
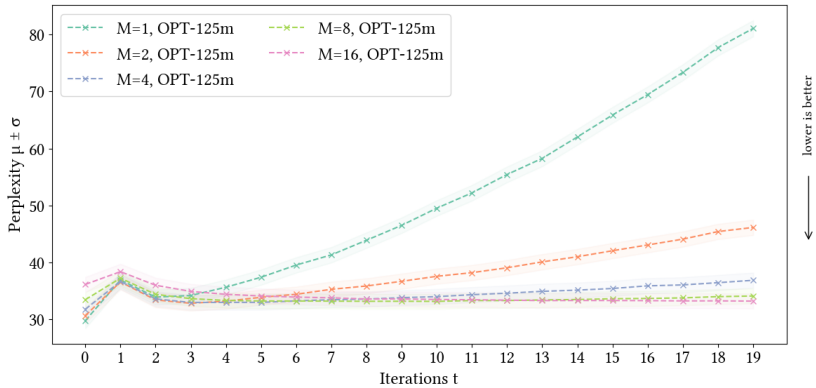


(a) Optimal level of diversity.



(b) Performance progress.

Fig. 10. **The optimal level of ecosystem diversity continues to increase beyond 10 self-training iterations (T).** (a) Optimal diversity level and the corresponding performance trajectories for OPT-125M. (a) and (b) extend the iteration range of Figures 3 and 4. Colored symbols correspond to the diversity levels shown in Figure 2 with an additional level of $D = 8$ in light green.

Diversity via data expansion. The expansion approach reflects a more realistic landscape of multiple similar models [40, 71]. If dominant models are trained on largely overlapping internet-scale data [47, 63], segmentation through reduction may be the only feasible diversification strategy; otherwise, diversity could be achieved via expansion by incorporating models trained on previously unused data. In such comparisons, diverse ecosystems consistently achieve lower perplexity across iterations (Figure 17b), whereas simply adding more similar models does not prevent collapse (Figure 20a).

Community-specific test sets. Evaluating community models using community-specific datasets rather than universal benchmarks yields a more realistic assessment; under this setting, diverse ecosystems outperform homogeneous ones already at $t = 0$ (Figure 17a). Community- or domain-specific models challenge the prevailing assumption that AI systems should serve everyone equally

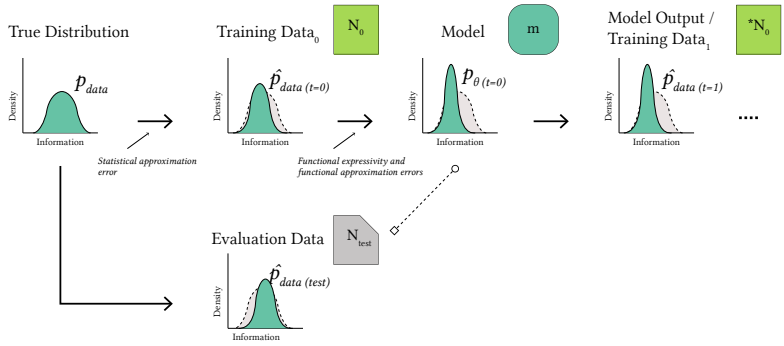


Fig. 11. **Illustration of probability distributions in self-training loops.** Because the training data is finite, the empirical distribution, $\hat{p}_{data(t=0)}$, diverges from the true distribution, p_{data}. The model's inherent biases (functional expressivity and functional approximation errors) further distort the distribution it represents, which is $p_{\theta(t=0)}$, and the distribution of the data it generates, which is $\hat{p}_{data(t=1)}$.

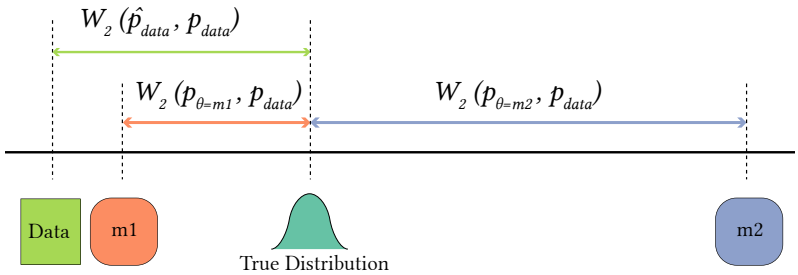


Fig. 12. **Illustration of model-dependent EDQ.** For the illustrated distribution discrepancies, the training data ("DATA") is beneficial for "m2" (high EDQ) but harmful for "m1" (low EDQ). EDQ is defined as $\delta_{EDQ}(\hat{p}data, p\theta) := \mathbb{W}2(\hat{p}data(t), p_{data}) - \mathbb{W}2(p\theta(t), p_{data})$, where lower values of δ_{EDQ} indicate higher EDQ. It is important to note that the introduced scalar metric serves as a heuristic to explain model-dependent data quality; therefore, we ignore additional factors shaping model weights.

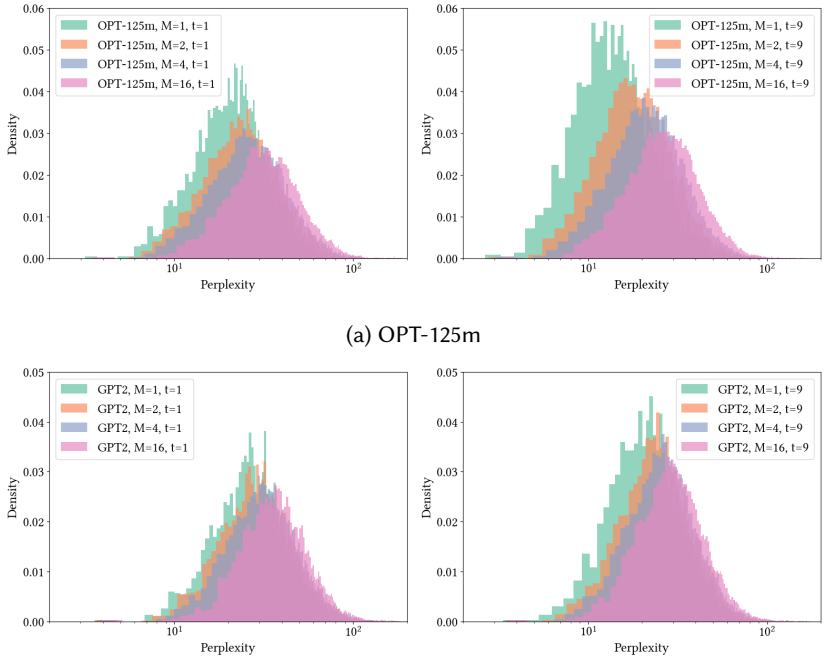
across all tasks [7, 9]. Granting affected communities greater agency in AI design [13] may naturally guide ecosystems toward healthier diversity.

Human language and knowledge evolve alongside AI systems. Following prior work [e.g., 1, 31, 56], we use a static dataset for generation and evaluation; in practice, this would resemble evaluating future models on today's data. A more realistic approach would involve evolving generation and evaluation datasets over time.

While further research is needed to identify optimal ecosystem parameters under realistic conditions, our results provide strong evidence that diversity is a key factor in sustaining long-term AI ecosystem performance.

H Sensitivity analysis

We conducted additional experiments to isolate factors that might affect the performance changes observed in the primary setting. The results indicate that neither (model-independent) data quality (Figure 18), training data size (Figure 19), the number of models per ecosystem (Figure 20), nor overfitting (Figure 21) can fully account for the observed performance decay or the differences



(a) OPT-125m

(b) GPT2

Fig. 13. **Perplexity distributions widen with increasing diversity.** Mixed perplexity distribution on the original Wikitext2 train data points, evaluated by all models in a given ecosystem with diversity ($D = M = 1, 2, 4, 16$) at iteration steps $t = 0$ (left) and $t = 9$ (right). As number of models increases, the distributions become wider, indicating greater diversity. Simultaneously, their means shift toward higher perplexity, suggesting greater divergence from the distribution in the original training set. Across all ecosystems, the distributions narrow from the beginning to the end of the model iterations.

across ecosystems. However, overfitting, which is captured by the EDQ, appears to be an important contributor to performance decay.

Because these individual factors fail to explain the performance trajectories in isolation, the findings provide support for the EDQ hypothesis.

I Hyperparameters

Table 1 provides the hyperparameters used for text generation.